

Performance evaluation of CPA firms in Taiwan from the perspective of industry-specific client groups

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Abstract This paper explores the operating efficiency of certified public accounting (CPA) firms from the perspective of industry-specific client groups. The decision making units are 49 partnership CPA firms selected from the database of 2010 Public Accounting Firms Service Investigation Report. This paper integrates three analysis methods: Data envelopment analysis, an independent sample t test, and multiple regression analysis. It aims to offer the operators of CPA firms with a reference for improving operating efficiency. It also identifies important industry-specific client groups for the sustainable operation of the firms, and analyzes the impact of operating efficiency on operating revenue and total revenue. It is believed that the results of this paper could provide decision references for future client relationship management and exploitation of new client groups.

Keywords Certified public accounting (CPA) firms · Operating efficiency · Industry-specific client groups · Data envelopment analysis (DEA) · Regression analysis

1 Introduction

The 2010 Public Accounting Firms Service Investigation Report, issued in November 2011 by the Financial Supervisory Commission of Taiwan, indicated that there were 954 certified public accounting (CPA) firms at the end of 2010, compared to 943 CPA firms at the end of 2009, for an increase of 1.2 % (see Panel A of Table 1); the number of CPA firms or workplaces clearly increased between 2007 and 2010. The Public Accounting Firms Service Investigation Report (2011)

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also indicated that between 2006 and 2010 there were 208 new CPA firms, comprising 21.9 % of the total number. There was an annual increase of 42 new CPA firms. It is apparent that as the numbers increase, CPA firms will face increasingly fierce competition. The 2010 Public Accounting Firms Service Investigation Report (see Panel B of Table 1) indicated that there were 697 CPA firms with sole proprietorship at the end of 2010. Of these, 257 CPA firms were operated by two practicing accountants. In terms of the number of employees (see Panel C of Table 1), 842 CPA firms, accounting for 88.3 %, had fewer than 20 employees; 16 CPA firms (1.7 %) employed over 100 people. The Big Four firms employ 41.6 % of all employees, in other words, nearly half.

Regarding the scale of operating revenue (see Panel D of Table 1), the Public Accounting Firms Service Investigation Report (2011) indicated that at the end of 2010, 554 CPA firms had annual operating revenue of less than NTD 5 million. These firms employed 12.9 % of the total employees in the industry and their operating revenue was 4.8 % of the CPA industry. There were 207 large CPA firms, comprising 21.7 % of the total, with an operating revenue exceeding NTD 10 million. The operating revenue comprised 89.4 % of the CPA industry. These companies employed 77.0 % of the employees in the industry.

It must be noted that the Big Four's operating revenue comprised 63.0 % of the CPA industry. While there are many small CPA firms, the operating effectiveness of large CPA firms means that they occupy a more important position and have more influence in the industry.

The first aim of this paper is to evaluate the operating performance of large CPA firms. Generally, the operating scale of the CPA firms that provide auditing and attestation services for the financial reports of public companies differs from those that do not provide such services. The CPA firms that provide attestation services for the financial reports of public companies are generally the larger firms.

In the first part of the study, the CPA firms providing attestation services for the financial reports of public companies are studied. There is a discussion of the indicators of CPA firms' operating efficiency; variation in returns to scale; improvement percentages of output and input and their contribution degree to the operating efficiency; how these are used to find causes of inefficiency; and identifying major industry-specific client groups in order to manage future client relationships.

The Public Accounting Firms Service Investigation Report (2011) gave a breakdown of industry-specific operating revenue (see Panel E of Table 1). The annual operating revenue of the CPA firms in 2010 indicated that revenue from the manufacturing industry was NTD 6.68 billion, accounting for 27.8 %; that from the information and communications industry was NTD 5.09 billion, accounting for 21.2 %; and that from the wholesale and retail industry was NTD 4.49 billion, accounting for 18.7 %. These three industry-specific client groups accounted for 67.7 % of the annual operating revenue, and were the main source of operating revenue. Due to the different characteristics of the industry, industry-specific client groups may lead to the differences in a firm's operating performance.

In the second part of this paper, CPA firms are divided into two groups: those that are efficient and those that are inefficient, based on overall and pure technical

Table 1 The overview of CPA industry

Panel A: The number of CPA firms and workplaces (Unit: number of firms)

	2007	2008	2009	2010
CPA firms	854	932	943	954
Workplaces	1,024	1,104	1,120	1,124

Panel B: The organization type of CPA firms (Unit: number of firms)

	2007	2008	2009	2010
Sole proprietorship	607	655	672	697
Co-operated by two practicing accountants	247	277	271	257

Panel C: The number of employees of CPA firms in 2010

The number of employees	Number of CPA firms
<5	348
5–9	295
10–19	199
20–29	39
30–39	24
40–49	13
50–99	20
100–199	11
200–299	–
>300	5
Total	954

Panel D: The scale of operating revenue of CPA firms in 2010

The scale of operating revenue	Number of CPA firms	Number of employees	Total operating revenue (Unit: NTD thousand)
<50 ten thousand	96	232	18,629
50 ten thousand~	71	172	54,071
100 ten thousand~	170	644	294,786
250 ten thousand~	217	1,379	788,161
500 ten thousand~	193	1,910	1,383,169
1,000 ten thousand~	133	2,251	1,899,290
2,500 ten thousand~	40	1,513	1,360,550
5,000 ten thousand~	18	1,250	1,230,146
>100 million	16	9,507	17,019,935
Total	954	18,858	24,048,737

Panel E: The percentage of operating revenue classified by industry categories in 2010

Industry categories	The percentage of operating revenue
Manufacturing	27.8

Table 1 continued

Panel E: The percentage of operating revenue classified by industry categories in 2010

Industry categories	The percentage of operating revenue
Information and communications	21.2
Wholesale and retail	18.7
Other industries	31.8
Personal business	0.5
Total	100

The above data from Panels A to E are summarized and reorganized according to the Public Accounting Firms Service Investigation Report in 2010 issued by the Financial Supervisory Commission of Taiwan, pp. 2, 3, 7–8, and 14

efficiency. The second aim of this paper is to analyze the differences in output and input in the two sample groups, as well as the differences in contribution the two sample groups made to operating efficiency, according to the degree of output and input. The research findings could help the operators of CPA firms to actively develop and maintain main industry-specific client groups, and determine applicable input items for efficient firms and inefficient firms.

The third aim of this paper is to develop multiple regression models to determine the correlation between the operating efficiency of CPA firms and their operating revenue and total revenue. Overall technical efficiency and pure technical efficiency served as the independent variables of the model; operating revenue and total revenue served as the dependent variables; and operating strategy and total number of employees served as the control variables. Using these variables, this paper constructs four multiple regression models to evaluate whether the operating efficiency of CPA firms is positively correlated with operating revenue and total revenue.

This paper integrates data envelopment analysis (DEA), an independent sample t test, and multiple regression analysis to identify important industry-specific client groups for CPA firms. The aim is to explore the effects of operating efficiency on operating revenue and total revenue, provide suggestions for the operators of CPA firms and help them make operating decisions. Moreover, in the business environment where the supply of services continuously increases, the research findings could help operators to find important industry-specific client groups that would be advantageous to their firms, provide references for future client relationship management and the development of new industry-specific client groups.

2 Literature review

2.1 Literature on different areas' applications of DEA

DEA is widely applied in performance evaluation. In *the high-tech industry*, Ma and Goo (2005) analyze technical efficiency and productivity changes in China's

high-tech and new technology industry development zones using DEA. Lai (2007) uses DEA to assess the relative efficiency of laptop computer manufacturers. Hung and Wang (2012) conduct DEA performance characterization of both high-tech and older industries. In *the CPA industry*, Lee (2009) applies DEA to analyze overall technical efficiency, pure technical efficiency, and scale efficiency in medium-size audit firms.

In *the banking industry*, Lin et al. (2009) use DEA to analyze a bank's operating performance. Yang (2009) uses DEA to assess the performance of the branches of Canadian banks. Köseadağ et al. (2011) apply DEA to investigate a possible convergence of the operational efficiency of financial institutions across countries. Paradi et al. (2011) use DEA to conduct a two-stage evaluation of efficiency in the branches of banks. George and Chattopadhyay (2012) use DEA to measure operational efficiency and performance benchmarking of Indian public sector banks. Minh et al. (2013) estimate and compare the efficiency performance of 32 commercial banks in Vietnam during 2001–2005 and identify possible factors determining such efficiency performance. In the education industry, Montoneri et al. (2011) use DEA to evaluate the learning performance of English writing courses in order to benchmark them. Montoneri et al. (2012) use DEA to assess the indicators that contribute to learning and teaching performance. Portela et al. (2012) describe a performance assessment of Portuguese secondary schools by using DEA.

In other areas, Despotis (2005) measures human development in Asia and the Pacific via DEA. Ramanathan and Yunfeng (2009) evaluate the incorporation cost and environmental factors in quality function deployment by using DEA. Lee (2010) uses DEA and decision trees for efficiency analysis and recommendation of B2C controls. Zhong et al. (2011) apply DEA to evaluate the performance of regional R&D investments in China. Huang et al. (2012) develop a performance evaluation model based on DEA to determine casino management efficiency in 2007 and the change in efficiency for eleven casinos in Atlantic City from 2003–2007. Huang and Kao (2012) assess the corporate environmental performance with the DEA method. Sahoo and Acharya (2012) construct the macroeconomic performance index of Indian states by using DEA.

2.2 Literature on the relationship between service quality and performance

The service industry has become more and more important for business activities (Tu et al. 2011). McKee (2008) indicates that the service industry contributes about 60 % of annual GDP and creates 70 % of new jobs in America. In addition, according to the statistics of Executive Yuan of Taiwan, the service industry contributed over 70 % of the annual GDP in 2008 (Tu et al. 2011). There are many studies that have to do with service quality and the factors which affect customer satisfaction and loyalty in various industries. The aim of these studies is to improve the performance of the service industry (e.g., Davis 1999; Santos 2003). Grönroos (1984) and Parasuraman et al. (1991) emphasize that perceptions of service quality have to do with the comparison of consumer expectations and actual performance. Ramayah et al. (2011) show that market orientation has a significant effect on

organizational performance and service quality. Besides, service quality has a significant effect on organizational performance.

Many studies indicate that better quality products and services relate to better organizational performance (e.g., Forker et al. 1996; Golhar and Deshpande 1999; Kroll et al. 1999). Taylor and Baker (1994) observe that service quality and customer satisfaction are closely related; service quality influences consumer satisfaction and vice versa. Jamali (2007) finds that internal performance, which is a reflection of actual service quality, needs to be enhanced to customers' satisfaction with service quality. Sedláček et al. (2011) analyze perception of quality as a factor of performance of companies operating in the tourism industry. Their results show that companies with higher levels of service quality achieve better financial indicators. Tu et al. (2011) find that service quality significantly affects customers' perception of value and customer satisfaction. These two aspects have a strong impact on customer loyalty.

2.3 Literature on the relationship between human resources and performance

Human capital includes: collective capabilities, experience, skills, leadership, intelligence, commitment, and how individual employees can solve problems (e.g., Bontis et al. 2000; Carson et al. 2004; Cohen and Kaimenakis 2007). Isaac et al. (2010) indicate that human capital represents the value added by employees when they utilize their knowledge and other personal assets to produce goods and services to create wealth for their organizations. Several previous studies have shown a positive association between effective human resource management (HRM) practices and organizational performance (e.g., Becker and Huselid 1998; Huselid 1995). Prowse and Prowse (2010) explore the theory that HRM could contribute to the improvement of organizational and individual performance.

Hitt et al. (2001) show that human capital has a positive effect on performance. Furthermore, their results show that human capital moderates the relationship between strategy and firm performance. Chen et al. (2008) show that the performance of audit firms with high levels of professional training and large manpower are significantly better than that of audit firms with low levels of professional training and less manpower. In addition, their results also indicate that both professional training and manpower have positive effects on performance. Chin and Sofian (2011) indicate that the existence of a close relationship between human capital and total quality management (TQM) and the integration of these has a positive impact on the corporate performance.

2.4 Literature on the relationship between information technology (IT) and performance

Al qeed et al. (2010) state that the tourism industry has widely adopted IT to reduce costs, enhance operational efficiency and, most importantly, to improve service quality and customer experience. Haberkamp et al. (2010) show that the impact of IT is greater on the strategic variable of productivity, followed by inter-organizational coordination. Hassandoust and Farzaneh (2011) maintain that IT

applications are not used merely to replace the paperwork and the traditional manual systems, but also to improve customer service and enhance management power. Evangelista et al. (2012) indicate a strong relationship among data gathering technologies, transactional capabilities and firm performance, in terms of both efficiency and effectiveness. Moreover, a positive correlation between enterprise IT and third-party logistics (3PL) financial performance has been found. Ringim et al. (2012) examine the critical success factors (CSFs) of business process management (BPM) and small to medium bank performance in the Nigerian banking industry. They identify five CSFs of BPM implementation: IT investment, volume of financial activities, personnel commitment, a strong capital base, and an effective reward system. Among these factors, IT investment, personnel commitment, and volume of financial activities have a significant relationship with overall organizational performance (cost reduction, customer service management, and operational efficiency performance).

2.5 Literature on the relationship between firm size and performance

Companies of different sizes generate different profits and operating performance; firm size is an important variable that affects operating performance. Thus, when analyzing the performance of the company, firm size should be included in the analysis (Gosselin 1997; Ittner et al. 2002). DeAngelo (1981) indicates that audit quality is positively correlated with the size of an accounting firm. Cheng et al. (2000) find that the larger the firm, the higher the technological efficiency. Lee et al. (2013) indicate that the scale of the customers' companies is positively correlated with the outsourcing probability of an express delivery company. Hung and Wang (2012) examine the managerial performance efficiency of 367 manufacturing firms in Taiwan which used two-stage DEA. Their empirical results indicate that the size of the firm plays an important role in influencing the operating efficiency. Lee (2012) uses the fixed asset value of a firm to measure the size of the accounting firm. The results show that firm size will influence the conduct and the performance of accounting firms.

3 Methodology

A three-stage study is designed to analyze CPA firms. In the first stage, DEA is used to evaluate various operating efficiency indicators of CPA firms, variation in returns to scale, and improvement percentages of output and input items, as well as their contribution to operating efficiency. The purpose of this stage is to determine the causes of inefficiency and to develop the management of client relationships for major industry-specific client groups.

In the second stage, decision making units (DMUs) are divided into two groups: efficient and inefficient groups according to overall technical efficiency and pure technical efficiency. This paper analyzes the difference between the output and input of the two sample groups and the difference in the degree of output and input in regard to operating efficiency. It is suggested that both the efficient and inefficient

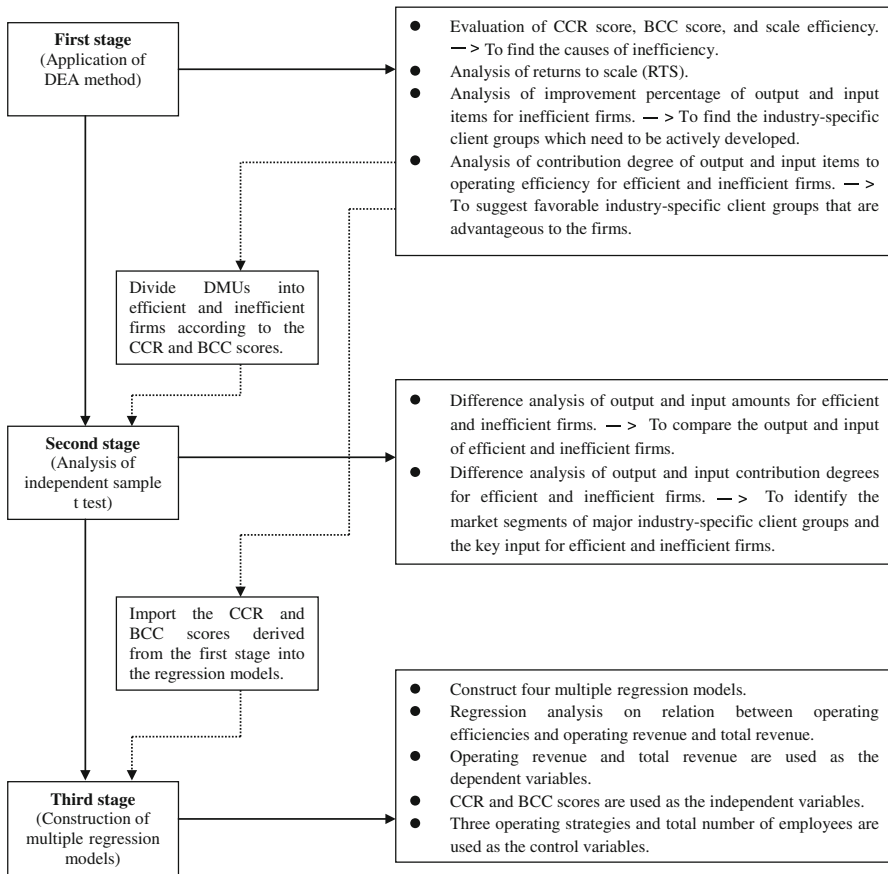


Fig. 1 The research framework diagram of this study

CPA firms can actively develop and maintain the major industry-specific client groups. Recommendations are made for suitable inputs.

In the third stage, this paper develops multiple regression empirical models for the correlation between operating efficiency and operating revenue (and total revenue). Overall technical efficiency and pure technology efficiency serve as the models' independent variables, operating revenue and total revenue serve as the dependent variables, and operating strategy and total number of employees serve as the control variables. Four multiple regression models are constructed to evaluate whether the operating efficiency of CPA firms have a positive impact on operating revenue and total revenue. The research framework is demonstrated as follows: Fig 1

3.1 Introduction of data envelopment analysis (DEA)

The DEA, proposed by Farrell (1957), is a production efficiency evaluation method in which the nonparametric production function, instead of the parametric function,

serves as the estimated efficiency value. Later, Charnes et al. (1978) extend the model to a multiple-input and multiple-output model, commonly referred to as the Charnes–Cooper–Rhodes (CCR) model, used to measure the overall technical efficiency of input and output under the constant returns to scale (CRS) assumption. Banker et al. (1984) change the CRS assumption premise of the CCR model to VRS (variable returns to scale), commonly referred to as the Banker–Charnes–Cooper (BCC) model and use this model to measure the pure technical efficiency. The scale efficiency is measured by the overall technical efficiency of the CCR model divided by the pure technical efficiency of the BCC model (Lee 2009; Lin et al. 2009). In this paper, the CCR model and BCC model of DEA are used to evaluate operating performance. An evaluation is conducted of overall technical efficiency, pure technical efficiency, and scale efficiency under the output-oriented premise; this is defined as the maximization of outputs with given inputs and in the CCR model is assumed to be CRS. For example, this means that double input leads to double output. In the BCC model, it is assumed to be VRS; for example, this means that double input leads to less than double output. This paper compares the samples of efficient and inefficient firms to analyze the differences in output and input in the two sample groups. It also analyzes the differences in contribution degree of output and input to operating efficiency, in order to identify the major profitable industry-specific client groups and key inputs.

3.2 Data sources, DMUs, and selection of output and input items

For the purposes of this paper, the DMUs are 49 partnership CPA firms selected from the database of the 2010 Public Accounting Firms Service Investigation Report. They provide auditing and attestation services for public companies' financial reports. The CPA industry provides services for 19 industries: agriculture, forestry, animal husbandry and fishery, mining and quarrying, manufacturing, power and gas supply, water supply and pollution control, construction, wholesale and retail, lodging, food and beverages, transport and storage, information and communications, banking and insurance, real estate, professional and technical services, support services, public administration and national defense, compulsory social security, education services, medical and healthcare, social work, arts, entertainment and leisure services, and other services.

When selecting the industry-specific client groups for analysis, the criterion is that the industries have more than two-thirds of their business transactions with the 49 firms in the current year. After selection, the industries with high homogeneity and those which are unpopular are eliminated. A total of six industry-specific client groups are selected as the basis of the firms' output sources. These industries include: manufacturing, wholesale and retail, lodging, food and beverages, transport and storage, information and communications, and professional and technical services industries. The firms' operating revenue from the six industries are the DEA model's six outputs (O1–O6).

As the CPA industry is a labor- and knowledge-intensive industry, the labor cost (I1) is used as an input. Moreover, as capital is one of the important inputs that sustain corporate operations, the operating assets cost (I2) is used as another input. The output and input items are defined in Table 2.

Table 2 Definitions of output and input items

Items	Names	Definitions
Outputs	Operating revenue from manufacturing industry (O1) (Unit: NTD)	It means annual operating revenue from manufacturing industry.
	Operating revenue from wholesale and retail industry (O2) (Unit: NTD)	It means annual operating revenue from wholesale and retail industry.
	Operating revenue from lodging, food and beverages industry (O3) (Unit: NTD)	It means annual operating revenue from lodging, food and beverages industry.
	Operating revenue from transport and storage industry (O4) (Unit: NTD)	It means annual operating revenue from transport and storage industry.
	Operating revenue from information and communications industry (O5) (Unit: NTD)	It means annual operating revenue from information and communications industry.
	Operating revenue from professional and technical services industry (O6) (Unit: NTD)	It means annual operating revenue from professional and technical services industry.
Inputs	Labor cost (I1) (Unit: NTD)	It includes salary, meals, employee benefits, provision for pensions, pension funds and overtime expenditures.
	Operating assets cost (I2) (Unit: NTD)	The sum of the costs from houses, buildings, ancillary equipment, furniture and fixtures, and transportation and other equipments.

3.3 Empirical regression models

Pandit et al. (2009) define surgical list “efficiency” as maximizing theatre utilization, minimizing over-running, and minimizing cancellations. “Efficiency” maximizes output for input; “productivity” emphasizes total output. Kumar and Gulati (2010) maintain that high efficiency does not stand for high effectiveness in the Indian public sector bank (PSBs) industry. A positive and strong correlation between effectiveness and performance measures has been noted. This paper further discusses whether operating efficiency affects operating revenue and total revenue; that is, whether the operating revenue and total revenue are higher when the operating efficiency is higher. Thus, CCR and BCC scores from DEA are used as the independent variables of the regression model, and operating revenue (PR) and total revenue (TR) are used as the dependent variables of the regression model.

In the database of the 2010 Public Accounting Firms Service Investigation Report, the questionnaires on future operating strategies of the CPA firms proposed eleven operating strategies, including improvement of service quality, recruitment of assistants, staff training, maintenance of present status, recruitment of accountants, business management informatization, expansion of business to mainland China, merging of other firms, establishment of management consulting firms, an increase of firm branches and others. Among the eleven operating strategies, the top three strategies that could increase PR and TR are used as the control variables of the regression model. The PR and TR of the firms may be affected by the size of the firm. Thus, the total number of employees is used as another control variable of the regression model. This paper conducts the ordinary least squares approach (OLS)

regression models, and develops four relational models (1)–(4) to do with the relation between the operating efficiencies and operating and total revenues of the CPA firms. The four models are listed as follows:

Model 1: The relation between operating revenue and CCR score

$$PR = a_0 + a_1CCREF + a_2ST1 + a_3ST2 + a_4ST3 + a_5EMP + e_1 \quad (1)$$

Model 2: The relation between total revenue and CCR score

$$TR = b_0 + b_1CCREF + b_2ST1 + b_3ST2 + b_4ST3 + b_5EMP + e_2 \quad (2)$$

Model 3: The relation between operating revenue and BCC score

$$PR = c_0 + c_1BCCEF + c_2ST1 + c_3ST2 + c_4ST3 + c_5EMP + e_3 \quad (3)$$

Model 4: The relation between total revenue and BCC score

$$TR = d_0 + d_1BCCEF + d_2ST1 + d_3ST2 + d_4ST3 + d_5EMP + e_4 \quad (4)$$

where PR is the operating revenue, including: revenue from attestation, taxation, management consulting, business registration, and other sources; TR is the total revenue, including: operating revenue, revenue from interest and rent, and other non-operating revenue; CCREF is the operating efficiency, derived from the CCR model with the DEA method; BCCEF is the operating efficiency, derived from the BCC model with the DEA method; ST1 is the binary variable, indicating whether the strategy of service quality improvement is adopted (ST1 = 1) or not (ST1 = 0); ST2 is the binary variable, indicating whether the strategy of recruitment of accountants is adopted (ST2 = 1) or not (ST2 = 0); ST3 is the binary variable, indicating whether the strategy of business management informatization is adopted (ST3 = 1) or not (ST3 = 0); EMP is the total number of employees; a_0, b_0, c_0, d_0 is the intercepts of the regression models; $a_1, a_2, a_3, a_4, a_5, b_1, b_2, b_3, b_4, b_5, c_1, c_2, c_3, c_4, c_5, d_1, d_2, d_3, d_4, d_5$ is the parameters of the regression models; e_1, e_2, e_3, e_4 is the error terms of the regression models.

4 Empirical results

4.1 Pearson correlation coefficient and descriptive statistics

After the Pearson correlation coefficient test has been completed, the results are computed. Table 3 shows that the correlation coefficients between output and input are higher than 0.7, indicating a significantly positive correlation, reaching the 1 %

Table 3 Pearson correlation coefficients between output and input items

	O1	O2	O3	O4	O5	O6
I1	0.977*** (0.000)	0.959*** (0.000)	0.831*** (0.000)	0.958*** (0.000)	0.997*** (0.000)	0.786*** (0.000)
I2	0.902*** (0.000)	0.921*** (0.000)	0.699*** (0.000)	0.989*** (0.000)	0.946*** (0.000)	0.721*** (0.000)

1. Output and input items are defined in Table 2. 2. The figures in () denote the p value. 3. The total DMUs are 49. 4. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

significant level. When input increases, output will also increase. The DEA empirical analysis could then be conducted.

Table 4 presents the descriptive statistics of output and input. With reference to the output, the average operating revenue from the information and communications industry (O5) is the highest (NTD 211,764,872), followed by the lodging, food and beverages industry (O3) (NTD 101,957,019). Third is the transport and storage industry (O4) (NTD 17,975,042). Of the six industry-specific client groups, the average operating revenue from the manufacturing industry (O1) is the lowest (NTD 2,170,848). It is apparent that the information and communications industry has gradually replaced traditional industries and has become the main and most important source of operating revenue.

With reference to the input, the average labor cost (I1) is NTD 101,402,025 and the average operating assets cost (I2) is NTD 46,396,530. The CPA firms belong to a labor-intensive industry and should provide high-quality professional services for clients; thus, human resource training should be the focus. Therefore, the human resource cost is a crucial input item for CPA firms.

4.2 Analysis of operating efficiency, improvement percentages, and contribution degree of output and input items

Table 5 shows the operating efficiency of the 49 CPA firms and improvement percentages of output and input. Among the firms, A45 and another 16 firms are operationally efficient; the overall technical efficiency (CCR score) is 1. Regardless of output or input, no improvement is needed; thus, the improvement percentage is 0. When the CCR score is 1, the pure technical efficiency (BCC score) and the scale efficiency are equal to 1. These firms have the status of being efficient, regardless of the configuration of output and input and the scale of the CPA firms' operations.

In terms of the number of references, A45 receives 25 and A49 receives 24. It is apparent that these firms are the benchmark; they are the firms that other, more inefficient firms most frequently imitated and learned from. Among the other 32 inefficient firms, the average CCR score is 0.740, the average BCC score is 0.802 and the average scale efficiency is 0.927, indicating that the inefficient firms should give priority to pure technical efficiency, improve the configuration of output and input, and then further improve the scale of operation, i.e., consider reduction or expansion of the scale of operation.

The returns to scale (RTS) under the BCC model show that A45 and another 24 firms are in the stage of constant returns to scale (CRTS); A20 and another five firms are in the stage of decreasing returns to scale (DRTS); and A2 and another 17 firms are in the stage of increasing returns to scale (IRTS). The number of firms with CRTS account for 51 % of the total.

An assessment of the average improvement percentage of output and input reveals that the most urgent output that needs to be improved is the operating revenue from the lodging, food and beverages industry (O3) which should be increased by 68,868,769.4 %. The next industry requiring improvement is the operating revenue from the information and communications industry (O5), which should be increased by 51,468,779.9 %. The operating revenue from the transport

Table 4 Descriptive statistics of output and input items

	O1	O2	O3	O4	O5	O6	I1	I2
Mean	2,170,848	8,827,966	101,957,019	17,975,042	211,764,872	8,834,351	101,402,025	46,396,530
Minimum	0	0	0	0	2,417,637	0	0	113,000
25th %	163,000	10,000	35,000	3,000	16,682,339	476,429	8,909,600	5,007,689
Median (50th %)	628,000	331,500	580,500	300,000	33,653,303	1,703,437	19,783,500	14,145,550
75th %	1,969,950	2,256,828	2,682,262	1,182,000	65,191,353	4,757,064	53,754,840	30,880,531
Maximum	24,128,659	141,268,299	2,199,174,967	399,273,844	3,750,844,819	117,600,607	1,549,746,325	752,674,118
Standard deviation	4,401,984	27,447,960	380,394,876	66,897,510	652,881,131	23,007,712	281,696,705	119,352,211

1. Output and input items are defined in Table 2. 2. The total DMU's are 49

Table 5 Analysis of operating efficiency of DMUs

DMUs	CCR score	DEA rankings	No. of references	No. of peers	BCC score	Scale efficiency	BCC RTS	Improvement percentage (Unit: %)							
								O1	O2	O3	O4	O5	O6	I1	I2
A45	1.000	1	25	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A49	1.000	1	24	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A23	1.000	1	17	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A18	1.000	1	13	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A27	1.000	1	12	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A41	1.000	1	10	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A30	1.000	1	9	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A22	1.000	1	7	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A12	1.000	1	6	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A25	1.000	1	6	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A21	1.000	1	4	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A8	1.000	1	3	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A39	1.000	1	3	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A46	1.000	1	3	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A38	1.000	1	2	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A5	1.000	1	1	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A36	1.000	1	1	0	1.000	1.000	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
A11	0.982	18	0	4	1.000	0.982	0	1.8	1.8	103.0	1,210.9	596.1	1.8	0.0	0.0
A34	0.970	19	0	6	1.000	0.970	0	3.1	3.1	1,758,835,228.8	3.1	3.1	3.1	0.0	0.0
A2	0.959	20	0	3	0.959	1.000	2	4.3	2,126.4	137.0	537.7	134.7	4.3	0.0	0.0
A40	0.956	21	0	4	1.000	0.956	0	4.6	4.6	13,517,253.3	1,085.6	39.7	4.6	0.0	0.0
A48	0.920	22	0	4	1.000	0.920	0	8.7	41.5	30.0	8.7	215.8	8.7	0.0	0.0
A14	0.892	23	0	5	1.000	0.892	0	12.2	12.2	12.2	276.4	291.3	12.2	0.0	0.0
A4	0.876	24	0	3	1.000	0.876	0	14.2	50.6	55.2	277.7	41.1	14.2	0.0	0.0
A26	0.843	25	0	4	0.925	0.911	2	18.6	18.6	42.8	298.5	439.7	18.6	0.0	0.0

Table 5 continued

DMUs	CCR score	DEA rankings	No. of references	No. of peers	BCC score	Scale efficiency	BCC RTS	Improvement percentage (Unit: %)											
								O1	O2	O3	O4	O5	O6	I1	I2				
A20	0.837	26	0	4	0.889	0.942	1	1,576.9	19.5	19.5	116,063,470.9	182,270,832.3	19.5	0.0	0.0				
A13	0.834	27	0	5	1.000	0.834	0	19.9	19.9	26,229,241.1	19.9	64,774,748.9	19.9	0.0	0.0				
A9	0.831	28	0	6	1.000	0.831	0	20.3	20.3	322.5	20.3	20.3	20.3	0.0	0.0				
A1	0.808	29	0	3	0.808	1.000	1	208.9	23.8	23.8	319,202,728.0	392.4	23.8	0.0	-82.2				
A44	0.794	30	0	4	0.998	0.795	2	26.0	76.7	26.0	26.0	270,030,243.8	111.3	0.0	0.0				
A3	0.773	31	0	5	0.775	0.998	1	29.4	29.4	29.4	105.9	122.5	29.4	0.0	0.0				
A31	0.760	32	0	4	0.918	0.828	2	31.6	31.6	101.0	559,533,145.2	31.6	1,560.6	0.0	0.0				
A10	0.754	33	0	3	0.763	0.989	2	32.7	32.7	45.6	483.0	63,417,904.2	34.9	0.0	0.0				
A43	0.741	34	0	3	0.822	0.902	1	764.2	34.9	44,357,456.0	29,562,356.9	92.7	39.3	0.0	0.0				
A35	0.718	35	0	4	0.837	0.858	2	39.3	39.3	258.8	6,698.3	496.5	39.3	0.0	0.0				
A37	0.687	36	0	6	0.688	0.999	1	45.5	45.5	45.5	912.7	45.5	45.5	0.0	0.0				
A16	0.682	37	0	4	0.731	0.933	2	46.6	46.6	76.7	207.3	625,468,129.1	46.6	0.0	0.0				
A24	0.682	38	0	4	0.688	0.990	2	46.7	46.7	284.2	181,133,439.0	105,689,751.4	46.7	0.0	0.0				
A29	0.661	39	0	4	0.793	0.833	2	51.4	51.4	62.6	104.9	51.4	12,621,892.0	0.0	0.0				
A42	0.660	40	0	3	0.699	0.944	2	51.6	51.6	51.6	838.6	71,302,525.6	81,785,957.2	0.0	-35.7				
A6	0.651	41	0	4	0.667	0.976	2	53.6	53.6	198.6	159,959,322.4	118,921,689.3	53.6	0.0	0.0				
A33	0.640	42	0	4	0.654	0.978	2	56.3	56.3	57.5	71.2	127.1	56.3	0.0	0.0				
A15	0.622	43	0	2	0.739	0.842	2	60.7	60.7	360,853,873.9	53,543,268.2	145,111,495.5	398.7	0.0	-45.0				
A7	0.606	44	0	4	0.610	0.993	2	65.0	65.0	147.2	448.6	65.0	4,315,279.7	0.0	0.0				
A28	0.600	45	0	3	0.642	0.934	2	66.7	66.7	1,157.5	2,330.4	883.8	4,332,915.1	0.0	0.0				
A47	0.512	46	0	5	0.515	0.993	2	95.5	95.5	106.1	465.6	95.5	95.5	0.0	0.0				
A32	0.484	47	0	4	0.506	0.957	2	106.6	106.6	390.2	1,281.1	9,152.1	106.6	0.0	0.0				
A19	0.484	48	0	4	0.493	0.981	1	106.8	106.8	3,667.2	19,718,834.0	154.8	106.8	0.0	0.0				
A17	0.463	49	0	4	0.550	0.841	2	116.0	116.0	116.0	3,045.8	144.0	107,527,263.5	0.0	0.0				

Table 5 continued

DMUs	CCR score	DEA rankings	No. of references	No. of peers	BCC score	Scale efficiency	BCC RTS	Improvement percentage (Unit: %)							
								O1	O2	O3	O4	O5	O6	I1	I2
Average of inefficient DMUs	0.740				0.802	0.927		118.3	111.1	68,868,769.4	44,960,541.3	51,468,779.9	6,692,833.6	0.0	-5.1

1. Output and input items are defined in Table 2. The figures in BCC RTS represent the returns to scale, "0" represents the constant returns to scale (CRTS), "1" represents the decreasing returns to scale (DRTS), "2" represents the increasing returns to scale (IRTS)

and storage industry (O4) should be increased by 44,960,541.3 %, and the operating revenue from the professional and technical services industry (O6) should be increased by 6,692,833.6 %. The improvement percentages from the six industries groups suggest that the CPA firms have more room for improvement in the development of these industry-specific client groups.

It is recommended that the CPA firms should actively develop these industry-specific client groups to improve operating efficiency, especially in the lodging, food and beverages industry, and the information and communications industry. The most urgent input that should be improved is that of the operating assets cost (I2) which should be slightly reduced by 5.1 %. However, there is no imperative to reduce the labor cost (I1). This fact reveals that labor cost is an important input item for the CPA industry.

Table 6 shows the contribution degree of output and input makes to operating efficiency. By observing the output of the 17 efficient firms, it is apparent that the average percentages of contribution degree of the operating revenue from the transport and storage (O4), manufacturing (O1), lodging, food and beverages industry (O3) rank as the top three for operating efficiency. The percentages of contribution degree are 26.4, 20.5, and 16.1 %, respectively; the percentages of the first two industries, in particular, exceed 20 %. It is recommended that, to promote operating performance, the operators of the efficient firms should focus on the development of the three industry-specific client groups and client relationship management. The average percentages of contribution degree of the labor cost (I1) and the operating assets cost (I2) to operating efficiency are 77.5 and 22.5 %, indicating that labor cost plays an important role in the CPA firms' operation and are an indispensable input item for maintaining the operations.

An assessment of the output of the inefficient 32 CPA firms shows that, in terms of operating efficiency, the average percentages of contribution degree of the operating revenue from the wholesale and retail industry (O2) and the manufacturing industry (O1) rank as the top two; 41.3 and 28.7 %, respectively. It is recommended that the operators of inefficient firms should focus on the development and maintenance of these two industries in order to improve the operating performance. The findings are consistent with those of the efficient firms. Regardless of whether the firms are efficient or inefficient, they focus on traditional clients and those in manufacturing. These traditional industries possess more large and major client groups in the market. Thus, these industry-specific client groups are important for the operating maintenance and increase of operating revenue of CPA firms. For input, the average percentages of contribution degree of the labor cost (I1) and the operating assets cost (I2) to operating efficiency are 92 and 8 %. The findings are consistent with those of the efficient CPA firms. This also reveals that the labor cost is an important input item of CPA firms' operations and plays a dominant role in business operations and service quality.

4.3 Difference analysis of amounts and contribution degrees from output and input items

According to the CCR and BCC scores, the CPA firms are divided into two groups: those that are efficient and those that are inefficient. As shown in Table 7, this paper

Table 6 Analysis of contribution degree of output and input items

DMUs	CCR score	DEA rankings	Contribution degree of output and input items (Unit: %)							
			O1	O2	O3	O4	O5	O6	I1	I2
A45	1.000	1	92.1	7.9	0	0	0	0	100	0
A49	1.000	1	0	0	0	100	0	0	6.5	93.5
A23	1.000	1	0	79.3	18.2	0	0	2.5	90.1	9.9
A18	1.000	1	42.2	29	0	0	0	28.8	99.9	0.1
A27	1.000	1	58.3	0	41.7	0	0	0	66.9	33.1
A41	1.000	1	14.6	1.9	0	21.1	27.1	35.4	85	15
A30	1.000	1	21	0	0	79	0	0	56.9	43.1
A22	1.000	1	0	0	0	0	100	0	100	0
A12	1.000	1	6.2	0	26.7	0	0.5	66.5	100	0
A25	1.000	1	2	86.3	11.7	0	0	0	81.4	18.6
A21	1.000	1	0	9.3	85.1	0	5.6	0	87.6	12.4
A8	1.000	1	21.8	0.7	0	28.9	30.3	18.3	72.8	27.2
A39	1.000	1	38.5	0.7	0	22.1	0	38.7	91.9	8.1
A46	1.000	1	0	0	0	100	0	0	0	100
A38	1.000	1	52.1	0	0.5	0.7	46.7	0	85.8	14.2
A5	1.000	1	0	8.6	90.5	0.9	0	0	96.8	3.2
A36	1.000	1	0	0	0	96.2	0	3.8	96.3	3.7
Average of efficient DMUs			20.5	13.2	16.1	26.4	12.4	11.4	77.5	22.5
A11	0.982	18	6.2	91.3	0	0	0	2.5	92.3	7.7
A34	0.970	19	8.7	4.8	0	40.6	35.5	10.5	64.5	35.5
A2	0.959	20	70.6	0	0	0	0	29.4	54.2	45.8
A40	0.956	21	7.6	84.3	0	0	0	8.1	93.5	6.5
A48	0.920	22	87.5	0	0	9.7	0	2.8	92.8	7.2
A14	0.892	23	15.2	62.7	11.1	0	0	11.1	94.8	5.2
A4	0.876	24	64.8	0	0	0	0	35.2	100	0
A26	0.843	25	6.3	41.7	0	0	0	52	97.9	2.1
A20	0.837	26	0	16.3	82.8	0	0	0.9	71.1	28.9
A13	0.834	27	5.6	20.7	0	13.2	0	60.4	89.9	10.1
A9	0.831	28	17.7	5.9	0	24.2	45.1	7.1	84.2	15.8
A1	0.808	29	0	5.8	87.8	0	0	6.4	100	0
A44	0.794	30	46.9	0	15.4	37.7	0	0	75.4	24.6
A3	0.773	31	10.7	42.1	23.7	0	0	23.5	86.4	13.6
A31	0.760	32	23.4	73.9	0	0	2.7	0	94.3	5.7
A10	0.754	33	25.4	74.6	0	0	0	0	93.9	6.1
A43	0.741	34	0	100	0	0	0	0	100	0
A35	0.718	35	55	21.6	0	0	0	23.4	99.6	0.4
A37	0.687	36	31.1	37.6	20.9	0	5.8	4.6	91.2	8.8
A16	0.682	37	7.8	52.5	0	0	0	39.7	97.4	2.6
A24	0.682	38	23.6	72.3	0	0	0	4.1	96.8	3.2

Table 6 continued

DMUs	CCR score	DEA rankings	Contribution degree of output and input items (Unit: %)							
			O1	O2	O3	O4	O5	O6	I1	I2
A29	0.661	39	19.9	67.8	0	0	12.3	0	98.5	1.5
A42	0.660	40	26.9	4.2	68.9	0	0	0	100	0
A6	0.651	41	30.2	64.1	0	0	0	5.7	95.7	4.3
A33	0.640	42	3	75.8	0	0	0	21.2	95.5	4.5
A15	0.622	43	95.2	4.8	0	0	0	0	100	0
A7	0.606	44	19.2	69.4	0	0	11.4	0	97.9	2.1
A28	0.600	45	29.1	70.9	0	0	0	0	93.3	6.7
A47	0.512	46	17.8	77.8	0	0	2.3	2.2	98.9	1.1
A32	0.484	47	61.8	31.2	0	0	0	7	99.2	0.8
A19	0.484	48	63.9	15.1	0	0	0	21	99.9	0.1
A17	0.463	49	38.1	33.7	28.2	0	0	0	95.9	4.1
Average of inefficient DMUs			28.7	41.3	10.6	3.9	3.6	11.8	92.0	8.0

Output and input items are defined in Table 2

analyzes the differences between output and input items in the efficient and inefficient samples. Panel A shows the results of the difference analysis of the CCR score division criteria. The results indicate that, except for the operating revenue from the wholesale and retail industry (O2) and the operating assets cost (I2), differences in the average values of output and input items of the two sample groups are positive. For the output, it is shown that the operating revenue of the efficient firms in five industries: manufacturing, lodging, food and beverages, transport and storage, information and communications, and professional and technical services, is higher than that of the inefficient firms. For the input, the labor cost of efficient firms is higher than that of inefficient firms, but all the differences in the average values fail to reach the significant level.

According to the difference analysis results of the BCC score division criteria, Panel B shows that the differences of average values of six outputs and two inputs are positive and reach the 5 % significant level. The operating revenue of the efficient firms is significantly higher than that of the inefficient firms. With reference to inputs, the labor cost (I1) and the operating assets cost (I2) of the efficient firms are also significantly higher than those of inefficient firms. Thus, the results of the BCC score division criterion are more consistent and more stable than those of the CCR score division criterion.

As shown in Table 7, the CPA firms are divided into efficient and inefficient samples according to the CCR and BCC scores in this paper. Table 8 focuses on the difference analysis in contribution degrees of output and input items to operating efficiency in the efficient and inefficient samples. According to the results of the difference analysis of the CCR score division criterion, Panel A shows that the

Table 7 Difference analyses in amounts of output and input items for efficient and inefficient DMUs

Output and Input items	CCR score	No. of DMUs	Average amount	Average difference	<i>t</i> value	<i>p</i> value (one-tailed)
Panel A: CCR score						
O1	Efficient	17	132,337,747	47,370,323	0.556	0.290
	Inefficient	32	84,967,424			
O2	Efficient	17	41,648,108	−7,271,021	−0.201	0.421
	Inefficient	32	48,919,129			
O3	Efficient	17	3,461,262	1,975,945	1.286	0.106
	Inefficient	32	1,485,317			
O4	Efficient	17	10,755,827	2,952,035	0.355	0.362
	Inefficient	32	7,803,792			
O5	Efficient	17	124,689,952	34,809,803	0.302	0.382
	Inefficient	32	89,880,150			
O6	Efficient	17	36,858,005	28,914,536	1.101	0.143
	Inefficient	32	7,943,470			
I1	Efficient	17	259,941,769	73,770,874	0.373	0.355
	Inefficient	32	186,170,895			
I2	Efficient	17	8,240,681	−909,059	−0.130	0.448
	Inefficient	32	9,149,740			
	BCC score	No. of DMUs	Average amount	Average difference	<i>t</i> value	<i>p</i> value (one-tailed)
Panel B: BCC score						
O1	Efficient	25	172,926,894	146,029,940	1.899**	0.035
	Inefficient	24	26,896,954			
O2	Efficient	25	76,032,290	60,506,344	1.854**	0.038
	Inefficient	24	15,525,947			
O3	Efficient	25	3,450,807	2,613,246	2.198**	0.019
	Inefficient	24	837,560			
O4	Efficient	25	16,811,430	16,299,570	2.201**	0.019
	Inefficient	24	511,860			
O5	Efficient	25	198,808,074	197,737,569	1.905**	0.034
	Inefficient	24	1,070,505			
O6	Efficient	25	34,275,195	33,279,478	1.817**	0.041
	Inefficient	24	995,718			
I1	Efficient	25	376,307,268	335,940,724	1.884**	0.036
	Inefficient	24	40,366,544			
I2	Efficient	25	14,600,236	11,772,013	1.872**	0.037
	Inefficient	24	2,828,223			

1. Output and input items are defined in Table 2. 2. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

differences in average contribution percentages of the operating revenue from the transport and storage industry (O4) and the operating assets cost (I2) to operating efficiency are significantly positive. This reveals that efficient firms are more

Table 8 Difference analyses in contribution degrees of output and input items for efficient and inefficient DMUs

Output and Input items	CCR score	No. of DMUs	Average contribution	Average difference	<i>t</i> value	<i>p</i> value (one-tailed)
Panel A: CCR score						
O1	Efficient	17	20.518	−8.207	−1.029	0.154
	Inefficient	32	28.725			
O2	Efficient	17	13.159	−28.182	−3.223***	0.001
	Inefficient	32	41.341			
O3	Efficient	17	16.141	5.554	0.711	0.240
	Inefficient	32	10.588			
O4	Efficient	17	26.406	22.487	2.285**	0.018
	Inefficient	32	3.919			
O5	Efficient	17	12.365	8.768	1.312	0.103
	Inefficient	32	3.597			
O6	Efficient	17	11.412	−0.426	−0.082	0.468
	Inefficient	32	11.838			
I1	Efficient	17	77.524	−14.508	−1.892**	0.037
	Inefficient	32	92.031			
I2	Efficient	17	22.476	14.508	1.892**	0.037
	Inefficient	32	7.969			
	BCC score	No. of DMUs	Average contribution	Average difference	<i>t</i> value	<i>p</i> value (one-tailed)
Panel B: BCC score						
O1	Efficient	25	22.484	−6.928	−0.910	0.184
	Inefficient	24	29.413			
O2	Efficient	25	19.736	−24.147	−2.697***	0.005
	Inefficient	24	43.883			
O3	Efficient	25	11.420	−2.234	−0.299	0.383
	Inefficient	24	13.654			
O4	Efficient	25	21.464	19.893	2.826***	0.004
	Inefficient	24	1.571			
O5	Efficient	25	11.632	10.195	2.105**	0.023
	Inefficient	24	1.438			
O6	Efficient	25	13.268	3.222	0.651	0.259
	Inefficient	24	10.046			
I1	Efficient	25	81.196	−11.846	−2.073**	0.023
	Inefficient	24	93.042			
I2	Efficient	25	18.804	11.846	2.073**	0.023
	Inefficient	24	6.958			

1. Output and input items are defined in Table 2. 2. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

dependent on the transport and storage industry than the inefficient firms are, and input more operating assets costs. However, the differences in average contribution percentages of the operating revenue from the wholesale and retail industry (O2) and the labor cost (I1) to operating efficiency are significantly negative. This suggests that inefficient firms are more dependent on the wholesale and retail industry and input more labor costs than efficient firms.

Based on the results of the difference analysis of the BCC score division criterion, Panel B shows that differences in average contribution percentages of the operating revenue from the transport and storage industry (O4), information and communications industry (O5), and the operating assets cost (I2) to operating efficiency are significantly positive. This reveals that efficient firms are more dependent on the transport and storage industry as well as the information and communications industry, and input more operating assets costs than inefficient firms. However, the differences in average contribution percentages of the operating revenue from the wholesale and retail industry (O2) and the labor cost (I1) to operating efficiency are significantly negative. This result is consistent with that of Panel A. As regards outputs, as discussed above, the efficient firms emphasize the operating revenue from the transport and storage industry, while the inefficient firms emphasize the operating revenue from the wholesale and retail industry. Regarding inputs, the efficient firms pay attention to the operating assets cost while the inefficient firms emphasize the labor cost. Thus, differences exist between the emphases that efficient and inefficient firms place on output and input items.

4.4 Regression results showing the relation between operating efficiency and operating revenue (and total revenue)

Table 9 shows the impact of CCR and BCC scores on operating revenue (PR) and total revenue (TR) in the four multiple regression models. An adjusted R^2 of the four models exceeds 0.9; the explanatory power of the models reaches the 1 % significant level and the fitness is better. The results from Model 1 to Model 4 indicate that both the CCR score (CCREF) and the BCC score (BCCEF) have significantly positive impacts on the two dependent variables, such as operating revenue (PR) and total revenue (TR), reaching a significant level of 10 %. In Model 3, the impact of the BCC score (BCCEF) on operating revenue (PR) reaches a significant level of 5 %. This shows that when the operating efficiency of the firms is better, the PR and TR are higher. In the control variables, ST1, ST2, and ST3 have no significantly positive impact on the two dependent variables, PR and TR. It can thus be assumed that the operating revenue and total revenue of the firms will not significantly change with an improvement in service quality, recruitment of accountants, and strategies for information for business management. The firms' total number of employees (EMP) has a significantly positive impact on two dependent variables, namely PR and TR, and reaches the 1 % significant level. In other words, with more employees, the scale of the operation is greater and the operating and total revenues are higher. The results are consistent with the expected results.

Table 9 Regression results on relation between operating efficiencies and operating revenue and total revenue

$PR = a_0 + a_1CCREF + a_2ST1 + a_3ST2 + a_4ST3 + a_5EMP + e_1$ (1)					
$TR = b_0 + b_1CCREF + b_2ST1 + b_3ST2 + b_4ST3 + b_5EMP + e_2$ (2)					
$PR = c_0 + c_1BCCEF + c_2ST1 + c_3ST2 + c_4ST3 + c_5EMP + e_3$ (3)					
$TR = d_0 + d_1BCCEF + d_2ST1 + d_3ST2 + d_4ST3 + d_5EMP + e_4$ (4)					
Variables	Predicted sign	Model 1:PR	Model 2:TR	Model 3:PR	Model 4:TR
Intercept		5.630*** (47.222)	5.640*** (47.610)	5.616*** (46.020)	5.625*** (46.411)
CCREF	+	0.190* (1.603)	0.182* (1.550)		
BCCEF	+			0.208** (1.685)	0.202* (1.647)
ST1	+	0.049 (0.961)	0.049 (0.981)	0.045 (0.900)	0.046 (0.924)
ST2	+	0.032 (0.441)	0.034 (0.469)	0.032 (0.441)	0.034 (0.472)
ST3	+	0.035 (0.383)	0.035 (0.392)	0.034 (0.376)	0.035 (0.388)
EMP	+	1.104*** (29.041)	1.105*** (29.236)	1.100*** (28.737)	1.101*** (28.944)
R^2		0.958	0.959	0.958	0.959
Adjusted R^2		0.953	0.954	0.954	0.954
F test		196.883***	199.469***	198.096***	200.892***
p value		<0.000	<0.000	<0.000	<0.000

1. Variables are defined in Section 3.3. 2. The figures in Table denote the coefficients. 3. The figures in parentheses denote the t values. 4. The variables' variance inflation factors (VIFs) are all less than 10, implying that no serious multi-collinearity exists among variables. 5. The total DMUs are 49. 6. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5 Discussion of the findings

The empirical results of this three-stage study indicate that there are 17 efficient firms among the 49 firms. Of the 32 inefficient firms, the average overall technical efficiency is 0.740, the average pure technical efficiency is 0.802, and the average scale efficiency is 0.927. The inefficient firms should first improve pure technical efficiency and the configuration between output and input. Then they should improve operating scale by considering reduction or expansion of scale. In addition, more than half of the firms reach the CRTS status.

According to average improvement percentages of output and input items, the first output which should be increased is the operating revenue from the lodging, food and beverages industry. Second to be improved should be the operating revenue from the information and communications industry. Regarding the

improvement percentages of outputs in the six industries, the firms have a larger improvement space to develop these industry-specific client groups. It is recommended that these industry-specific client groups should be actively developed to improve operating efficiency. The most urgent input which should be improved is the operating assets cost. However, labor cost should not be reduced.

Regarding the contribution degree of output and input items to operating efficiency, the operating revenues from the transport and storage, manufacturing, lodging, food and beverages industries in the efficient firms are ranked as the top three. It is recommended that the operators of efficient firms should focus on development of the three industry-specific client groups and manage client relationships better, in order to increase operating efficiency. In the inefficient firms, the operating revenues from the wholesale and retail and the manufacturing industries indicate that these are ranked as the top two. It is recommended that the operators of the inefficient firms focus on development of the two industry-specific client groups.

Regardless of whether a firm is efficient or inefficient, the focus of its business services should always be on developing the client group. The client group plays an important role in the maintenance of the CPA firm's operation and increase in operating revenue. Regardless of whether a firm is efficient or inefficient, labor cost plays an important role in the CPA firm's operation, and is an indispensable factor in service quality.

The difference analysis results of output and input items reveal that the operating revenues from all the industry-specific client groups in the efficient firms are significantly higher than those of inefficient firms. In the inputs, the labor cost and the operating assets cost of the efficient firms are also significantly higher than those of the inefficient firms. Thus, the results of the BCC score division criterion are more consistent and more stable than that of the CCR score division criterion.

According to the difference analysis of contribution degree of output and input items to operating efficiency, the efficient firms focus on the operating revenue from the transport and storage industry for their outputs, while the inefficient firms focus on the operating revenue from the wholesale and retail industry. For inputs, the efficient firms emphasize the application of operating assets cost, while the inefficient firms emphasize the application of labor cost. Hence, this paper suggests that different CPA firms should focus on different industry-specific client groups and inputs.

Finally, the regression results show that the overall technical efficiency and the pure technical efficiency have a significantly positive impact on operating revenue and total revenue. In other words, when the operating efficiency of the firms is higher, the operating revenue and total revenue are also higher. Among the control variables, the total number of employees has a significantly positive impact on operating revenue and total revenue. When the scale of the operation is larger, the operating revenue and total revenue are higher.

6 Conclusions and suggestions

This paper concludes that, first, the information and communications industry has become the most important source of operating revenue and that the cost of human

resources is a crucial input item for the CPA firms. Second, the research findings show that more than half of the CPA firms are inefficient. Thus, this paper suggests that the operators of CPA firms could improve the configuration between output and input items.

Third, this paper suggests that the CPA firms should actively develop clients from the lodging, food and beverages industry, and the information and communications industry to improve operating efficiency. In addition, because labor cost is an important input item for the CPA industry, it is not necessary to reduce the expenditure on this item. Fourth, it is recommended that the efficient CPA firms could actively develop management of their client relationships in the transport and storage industry and the manufacturing industry. The inefficient CPA firms could develop the management of their client relationships in the wholesale and retail industry and the manufacturing industry.

Finally, this paper identifies the important industry-specific client groups that are advantageous to the CPA firms with different operating efficiencies, in order to manage and maintain long-term client relationships. It discusses the impact of operating efficiency on operating revenue and total revenue. The findings provide useful references for CPA firms' operators in their decision making and help them to identify the market segments of the major client groups. Furthermore, this paper provides CPA firms' operators directions for future client relationship management and market development of new industry-specific client groups. These are the important contributions to the practitioners, as well as that to the academic body of knowledge of this paper.

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